Experiment 25

1. Title

Design and Development of an On-Road Vehicle Breakdown Assistance System using HTML, CSS and JavaScript.

2. Aim

To develop a web-based roadside assistance application that helps vehicle owners search for nearby or location-specific mechanics, view mechanic location on a map, and provide feedback. The system allows mechanics to register and submit business details, while an administrator reviews and approves mechanic listings before they become visible to users.

3. Software Requirements

Component	Requirement
Frontend	HTML5
Styling	CSS3
Client-side scripting	JavaScript (ES6)
Code editor	Visual Studio Code / any text editor
Browser	Google Chrome / Microsoft Edge / Firefox
Data storage	Browser LocalStorage (lab demonstration)
Map / geolocation	Google Maps link and browser Geolocation API
Operating system	Windows / Linux / macOS

4. Problem Description

A vehicle may stop working unexpectedly because of a flat tyre, battery failure, engine trouble or other mechanical problems. In an unfamiliar area, the driver may not know which mechanic to contact. Searching manually can waste valuable time, especially during a roadside emergency.

The On-Road Vehicle Breakdown Assistance System provides a common platform to connect vehicle owners with mechanics. Mechanics submit their registration and business information. The administrator checks the submitted information and approves eligible mechanics. Only approved mechanic listings are displayed in user searches. Users can search by location, open a map location, view their profile and post feedback.

The system has three roles:

User: registers, logs in, searches for approved mechanics, views map locations, posts feedback and manages a basic profile.

Mechanic: registers, logs in, posts business details, manages their business information and views feedback.

Admin: reviews mechanic applications and approves or rejects submitted mechanic registrations.

This lab implementation is a front-end prototype. LocalStorage is used for demonstration data. Actual licence verification, secure authentication, live GPS distance calculation, notifications and a production database require a trusted backend and external services.

5. Modules

5.1 User Modules

User Registration

User Login

Search Approved Mechanic Details by location

View Google Maps / geolocation link

Post Feedback

My Profile

5.2 Admin Modules

Admin Login

Review mechanic registration and business details

Provide approval or reject mechanic applications

5.3 Mechanic Modules

Mechanic Registration

Mechanic Login

Post Business Details

Manage Business Details

View Feedback

6. Algorithm

Start the application by opening index.html in a web browser.

Initialize sample users, mechanic records and feedback in LocalStorage if no data exists.

Display the home page with a mechanic search field and navigation menu.

Allow a user or mechanic to register with the required details.

Validate login credentials and identify the account role.

Allow a mechanic to submit business name, service location, contact number, services and licence number.

Store new mechanic applications with Pending approval status.

Allow the administrator to review pending applications and approve or reject them.

Display only approved mechanics when a user searches by location.

Allow the user to open the mechanic's Google Maps search link or request browser location permission.

Allow a logged-in user to post feedback about a mechanic.

Allow a mechanic to view feedback and update their own business details.

Stop when the user closes the application.

7. Complete HTML, CSS and JavaScript Code

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>RoadRescue - Vehicle Breakdown Assistance</title>

<style>

:root{--navy:#12344a;--blue:#147da3;--pale:#eaf3f7;--ink:#263b49}

*{box-sizing:border-box}

body{margin:0;font-family:Arial,sans-serif;background:#f4f8fa;color:var(--ink)}

header{background:var(--navy);color:white;padding:17px 5%;display:flex;

justify-content:space-between;align-items:center;gap:12px;flex-wrap:wrap}

header h1{font-size:23px;margin:0}

nav{display:flex;gap:7px;flex-wrap:wrap}
```

```
nav button{background:transparent;color:white;border:0;cursor:pointer;padding:8px}
main{max-width:1100px;margin:25px auto;padding:0 18px}
.hero{background:linear-gradient(120deg,#dcecf4,#c5e1eb);padding:42px 28px;
border-radius:15px;margin-bottom:24px}
.hero h2{font-size:32px;color:var(--navy);margin:0 0 12px}
.panel,.card{background:white;border:1px solid #d9e4ea;border-radius:12px;
padding:18px;box-shadow:0 3px 12px #12344a0b}
.panel{margin:14px 0}
.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(230px,1fr));gap:16px}
.card h3{margin:0 0 10px;color:var(--navy)}
button,.btn{background:var(--blue);color:white;border:0;border-radius:7px;
padding:10px 14px;cursor:pointer}
button:hover,.btn:hover{background:#0e6384}
input,textarea,select{width:100%;padding:10px;margin:6px 0 12px;
border:1px solid #cbd8df;border-radius:6px;font:inherit}
label{font-size:14px;font-weight:bold}
form{max-width:560px}
.hidden{display:none!important}
.notice{padding:11px;background:#e5f4e9;border-radius:7px;margin:12px 0}
.pending{color:#9a6400;font-weight:bold}.approved{color:#14744b;font-weight:bold}
.muted{color:#617583;font-size:14px}
table{width:100%;border-collapse:collapse;background:white}
td,th{border:1px solid #dbe4e9;padding:9px;text-align:left}
th{background:#eaf2f6}
footer{text-align:center;background:var(--navy);color:white;padding:18px;margin-top:35px}
@media(max-width:600px){.hero h2{font-size:25px}}
</style>
```

```
</head>

<body>

<header>

<h1>🛠 RoadRescue</h1>

<nav>

<button onclick="showPage('home')">Home</button>

<button onclick="showPage('search')">Find Mechanic</button>

<button onclick="showPage('register')">Register</button>

<button onclick="showPage('login')">Login</button>

<button onclick="showPage('profile')">My Profile</button>

<button onclick="logout()">Logout</button>

</nav>

</header>

<main>

<div id="message" aria-live="polite"></div>

<section id="home" class="page">

<div class="hero">

<h2>Help is closer than you think.</h2>

<p>Find approved mechanics for unexpected vehicle breakdowns. Search by city or area and get roadside help.</p>

<button onclick="showPage('search')">Find a Mechanic</button>

</div>

<div class="grid">

<article class="card"><h3>Verified listings</h3><p>Only mechanics approved by the administrator appear in search results.</p></article>

<article class="card"><h3>Location support</h3><p>Open a mechanic's location using Google Maps.</p></article>
```

```
<article class="card"><h3>Customer feedback</h3><p>Share feedback about the service received.</p></article>

</div>

</section>

<section id="register" class="page hidden">

<div class="panel"><h2>Registration</h2>

<form id="registerForm">

<label>Account type</label><select id="regRole"><option value="user">User</option><option value="mechanic">Mechanic</option></select>

<label>Full name</label><input id="regName" required>

<label>Email</label><input id="regEmail" type="email" required>

<label>Phone number</label><input id="regPhone" required>

<label>Password</label><input id="regPassword" type="password" minlength="4" required>

<button type="submit">Create Account</button>

</form>

</div>

</section>

<section id="login" class="page hidden">

<div class="panel"><h2>Login</h2>

<form id="loginForm">

<label>Email</label><input id="loginEmail" type="email" required>

<label>Password</label><input id="loginPassword" type="password" required>

<button type="submit">Login</button>

</form>

<p class="muted">Demo admin: admin@roadrescue.com / admin123</p>

</div>

</section>

<section id="search" class="page hidden">
```

```
<h2>Search Approved Mechanics</h2>

<div class="panel">

<label>Enter city / area</label><input id="searchLocation" placeholder="Example: Bengaluru">

<button onclick="searchMechanics()">Search Mechanics</button>

<button onclick="useMyLocation()">Use My Location</button>

<p class="muted">Search results contain approved mechanic listings only.</p>

</div>

<div id="mechanicResults" class="grid"></div>

</section>

<section id="profile" class="page hidden">

<div class="panel"><h2>My Profile</h2><div id="profileDetails">Please log in to view your profile.</div></div>

</section>

<section id="business" class="page hidden">

<div class="panel"><h2>Post / Manage Business</h2>

<form id="businessForm">

<label>Business name</label><input id="bizName" required>

<label>Service location (city / area)</label><input id="bizLocation" required>

<label>Contact number</label><input id="bizPhone" required>

<label>Services offered</label><textarea id="bizServices" placeholder="Car repair, battery, tyre, towing..." required></textarea>

<label>Mechanic licence number</label><input id="bizLicence" required>

<button type="submit">Submit Business for Approval</button>

</form>

<p class="muted">New or edited business details must be approved by the administrator before appearing in public search.</p>

<div id="myBusiness"></div>

</div>
```

```
</section>

<section id="feedback" class="page hidden">

<div class="panel"><h2>Post Feedback</h2>

<form id="feedbackForm">

<label>Mechanic</label><select id="feedbackMechanic" required></select>

<label>Rating</label><select
id="rating"><option>5</option><option>4</option><option>3</option><option>2</option><option>1
</option></select>

<label>Comments</label><textarea id="comments" required></textarea>

<button type="submit">Submit Feedback</button>

</form>

</div>

</section>

<section id="mechanicFeedback" class="page hidden">

<div class="panel"><h2>Customer Feedback</h2><div id="feedbackList"></div></div>

</section>

<section id="admin" class="page hidden">

<h2>Admin – Mechanic Approval</h2>

<div class="panel"><div id="adminList"></div></div>

</section>

</main>

<footer>© 2026 RoadRescue | Reliable Roadside Assistance</footer>

<script>

const initial={

users:[],

mechanics:[

{id:1,owner:"Ravi Kumar",email:"ravi@example.com",business:"Ravi Auto Works",

location:"Bengaluru",phone:"9876543210",services:"Car repair, battery replacement",
```

```
licence:"KA-MECH-101",status:"Approved"},

{id:2,owner:"Suresh",email:"suresh@example.com",business:"City Roadside Care",

location:"Mysuru",phone:"9876501234",services:"Tyre repair, towing, car service",

licence:"KA-MECH-102",status:"Approved"}

],

feedback:[]

};

function get(k){return JSON.parse(localStorage.getItem(k)||"null");}

function set(k,v){localStorage.setItem(k,JSON.stringify(v));}

function init(){Object.keys(initial).forEach(k=>{if(!get(k))set(k,initial[k]);});}

function esc(s){return String(s).replace(/[&<>"']/g,c=>({'&':'&amp;','<':'&lt;','>':'&gt;','"':'&quot;',"'":'&#39;'}[c]));}

function msg(t){document.getElementById("message").innerHTML='<div class="notice">'+esc(t)+'</div>';}

function current(){return JSON.parse(sessionStorage.getItem("account")||"null");}

function showPage(id){

document.querySelectorAll(".page").forEach(x=>x.classList.add("hidden"));

if(id==="admin"&&sessionStorage.getItem("role")!=="admin"){msg("Admin login required.");id="login";}

if(["business","mechanicFeedback"].includes(id)&&sessionStorage.getItem("role")!=="mechanic"){msg("Mechanic login required.");id="login";}

if(["profile","feedback"].includes(id)&&!current()){msg("Please log in first.");id="login";}

document.getElementById(id).classList.remove("hidden");

refresh();

}

function logout(){sessionStorage.clear();msg("Logged out successfully.");showPage("home");}

function refresh(){

const acc=current();

if(acc){
```

```
document.getElementById("profileDetails").innerHTML="<p><b>Name:</b> "+esc(acc.name)+
"</p><p><b>Email:</b> "+esc(acc.email)+"</p><p><b>Phone:</b> "+esc(acc.phone)+
"</p><p><b>Role:</b> "+esc(acc.role)+"</p>";
}else document.getElementById("profileDetails").textContent="Please log in to view your profile.";
const me=acc&&acc.role==="mechanic"?get("mechanics").find(m=>m.email===acc.email):null;
if(me){
document.getElementById("myBusiness").innerHTML="<p><b>Business:</b> "+esc(me.business)+
" — <span class='"+(me.status==="Approved"?"approved":"pending")+"'>"+esc(me.status)+"</span></p>";
document.getElementById("feedbackList").innerHTML=feedbackFor(me.email);
}
const approved=get("mechanics").filter(m=>m.status==="Approved");
document.getElementById("feedbackMechanic").innerHTML=approved.map(m=>
'<option value="'+m.id+'">'+esc(m.business)+'</option>').join("");
renderAdmin();
}
function feedbackFor(email){
const list=get("feedback").filter(f=>f.mechanicEmail===email);
return list.length?list.map(f=>"<p>⭐ "+esc(f.rating)+"/5 — "+esc(f.comment)+"</p>").join(""):"<p>No feedback yet.</p>";
}
document.getElementById("registerForm").addEventListener("submit",e=>{
e.preventDefault();const users=get("users"),email=regEmail.value.trim().toLowerCase();
if(users.some(u=>u.email===email)||get("mechanics").some(m=>m.email===email)){msg("Email already registered.");return;}
const role=regRole.value;
users.push({name:regName.value.trim(),email,phone:regPhone.value.trim(),
password:regPassword.value,role});set("users",users);
```

```
if(role==="mechanic"){
const mechanics=get("mechanics");mechanics.push({id:Date.now(),owner:regName.value.trim(),
email,phone:regPhone.value.trim(),business:"",location:"",services:"",licence:"",status:"Profile
Pending"});
set("mechanics",mechanics);
}
e.target.reset();msg("Registration successful. Please log in.");showPage("login");
});
document.getElementById("loginForm").addEventListener("submit",e=>{
e.preventDefault();const email=loginEmail.value.trim().toLowerCase(),password=loginPassword.value;
if(email==="admin@roadrescue.com"&&password==="admin123"){
sessionStorage.setItem("role","admin");msg("Admin login successful.");showPage("admin");return;
}
const user=get("users").find(u=>u.email===email&&u.password===password);
if(!user){msg("Invalid login details.");return;}
sessionStorage.setItem("account",JSON.stringify(user));sessionStorage.setItem("role",user.role);
msg("Welcome, "+user.name+".");
showPage(user.role==="mechanic"?"business":"home");
});
function searchMechanics(){
const q=document.getElementById("searchLocation").value.trim().toLowerCase();
const matches=get("mechanics").filter(m=>m.status==="Approved"&&
(!q||m.location.toLowerCase().includes(q)));
document.getElementById("mechanicResults").innerHTML=matches.length?matches.map(m=>
'<article class="card"><h3>'+esc(m.business)+'</h3><p><b>Mechanic:</b> '+esc(m.owner)+
'</p><p><b>Location:</b> '+esc(m.location)+'</p><p><b>Phone:</b> '+esc(m.phone)+
'</p><p><b>Services:</b> '+esc(m.services)+'</p><p class="approved">Approved mechanic</p>'+
```

```
'<a target="_blank" rel="noopener" href="https://www.google.com/maps/search/?api=1&query='+

encodeURIComponent(m.business+" "+m.location)+'">View Google Maps location</a></article>').join("")

:"<p>No approved mechanics found for this location.</p>";

}

function useMyLocation(){

if(!navigator.geolocation){msg("Geolocation is not supported by this browser.");return;}

navigator.geolocation.getCurrentPosition(pos=>{

const
url="https://www.google.com/maps/search/mechanics/@"+pos.coords.latitude+","+pos.coords.longitude
+",14z";

window.open(url,"_blank","noopener");msg("Your location was found. Google Maps has been opened.");

},()=>msg("Location permission was denied or location is unavailable."));

}

document.getElementById("businessForm").addEventListener("submit",e=>{

e.preventDefault();const acc=current();if(!acc||acc.role!=="mechanic"){msg("Mechanic login required.");return;}

const list=get("mechanics"),m=list.find(x=>x.email===acc.email);

if(!m){msg("Mechanic profile not found.");return;}

Object.assign(m,{business:bizName.value.trim(),location:bizLocation.value.trim(),

phone:bizPhone.value.trim(),services:bizServices.value.trim(),licence:bizLicence.value.trim(),

status:"Pending"});

set("mechanics",list);msg("Business submitted. It will appear in search after admin approval.");refresh();

});

function renderAdmin(){

const list=get("mechanics");

document.getElementById("adminList").innerHTML=list.length?
```

```
'<table><tr><th>Business / Owner</th><th>Location</th><th>Licence</th><th>Status</th><th>Action</th></tr>'+
list.map(m=>'<tr><td>'+esc(m.business||"Not submitted")+'<br>'+esc(m.owner)+'</td><td>'+
esc(m.location||"-")+'</td><td>'+esc(m.licence||"-")+'</td><td>'+esc(m.status)+'</td><td>'+
(m.status!=="Approved"&&m.business?'<button onclick="approve('+m.id+')">Approve</button> ':"")+
(m.status!=="Rejected"?'<button onclick="reject('+m.id+')">Reject</button>':"")+
'</td></tr>').join("")+'</table>':"<p>No mechanic applications.</p>";
}
function approve(id){const a=get("mechanics"),m=a.find(x=>x.id===id);if(m){m.status="Approved";set("mechanics",a);renderAdmin();msg("Mechanic approved and now eligible for search.");}}
function reject(id){const a=get("mechanics"),m=a.find(x=>x.id===id);if(m){m.status="Rejected";set("mechanics",a);renderAdmin();msg("Mechanic application rejected.");}}
document.getElementById("feedbackForm").addEventListener("submit",e=>{
e.preventDefault();const acc=current();if(!acc){msg("Please log in.");return;}
const m=get("mechanics").find(x=>x.id===Number(feedbackMechanic.value));
if(!m){msg("Select a valid mechanic.");return;}
const list=get("feedback");list.push({user:acc.name,mechanicEmail:m.email,
rating:rating.value,comment:comments.value.trim()});set("feedback",list);
e.target.reset();msg("Thank you. Your feedback has been submitted.");refresh();
});
init();refresh();
</script>
</body>
</html>
```

8. Sample Input

Use the following sample values while testing the application.

Operation	Field	Sample value
Mechanic registration	Name / Email / Phone / Password	Ravi Kumar / ravi2@example.com / 9876543211 / mech123
Mechanic business	Business / Location / Services	Ravi Car Care / Bengaluru / Car repair, battery service
Licence	Licence number	KA-MECH-205
Admin login	Email / Password	admin@roadrescue.com / admin123
Mechanic search	Location	Bengaluru
Feedback	Rating / Comment	5 / Quick and helpful roadside service

9. Expected Output with Image

When index.html is opened, the home page displays the RoadRescue title, navigation menu and a roadside assistance banner. Users can register and log in, search approved mechanics by city or area, and open a Google Maps search for a mechanic. Mechanics can submit business information, which remains pending until admin approval. The admin dashboard provides approval and rejection controls. Logged-in users can submit feedback.

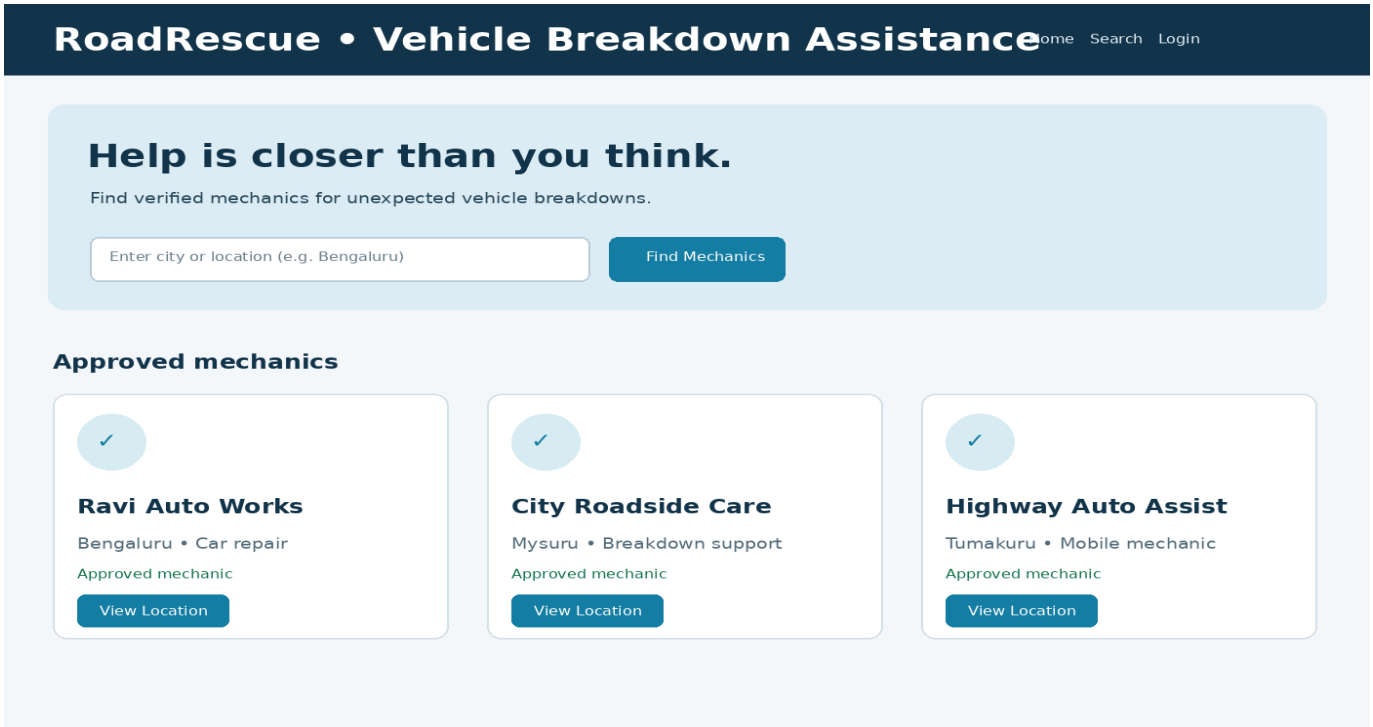


Figure 1: Illustrative expected home page and approved mechanic search results

10. Result:Thus, the On-Road Vehicle Breakdown Assistance System was designed and developed successfully using HTML, CSS and JavaScript. The prototype demonstrates user and mechanic registration/login, mechanic business submission, administrator approval, location-based search of approved mechanics, Google Maps location links, user feedback and profile viewing.

EXPERIMENT 26

1. Title
Traffic Squad: Penalty Collection and Management-Based Web Application

2. Aim
To develop a web application that enables traffic police to record traffic violations, attach photographic evidence, issue digital challans and allow vehicle owners to view cases and fine details. The administrator can manage police accounts and view daily reports.

3. Software Requirements

Component	Requirement
Frontend	HTML5
Styling	CSS3
Programming	JavaScript (ES6)
Editor	Visual Studio Code / text editor
Browser	Google Chrome / Microsoft Edge / Firefox
Storage	Browser LocalStorage (prototype only)
OS	Windows / Linux / macOS

4. Problem Description
Traffic violations are often recorded manually, making it difficult to maintain evidence, issue receipts promptly and track penalty collection. This system provides a centralized digital workflow. Police can select a violation rule, enter vehicle and owner details, upload a proof photo and create a case. Users can log in to view challans and cases. Admin can manage police access and view daily case reports.

5. Modules

User
- Register
- Login
- View Challan
- View Case Filed

Police
- Login
- View Traffic Rules
- Create Case

- Upload Proof Photo
- Manage traffic reports and new updates
- Open Google Maps traffic link

Admin

- Manage Police Login
- View Daily Report

6. Algorithm

1. Open the application in a browser and initialize sample traffic rules and accounts.
2. Register a vehicle owner account or log in with an existing account.
3. Police logs in and selects a traffic rule, then enters vehicle number, owner details, location and fine amount.
4. Police attaches a proof photograph and submits the case.
5. The system generates a challan reference and stores the case in browser storage.
6. The vehicle owner logs in and views challans and filed cases associated with their registered email or vehicle number.
7. Admin views the daily report and manages police account records.
8. End the session by logging out.

7. Complete HTML, CSS and JavaScript Code

```
<!DOCTYPE html>
<html lang="en"><head><meta charset="UTF-8"><meta name="viewport" content="width=device-width,initial-scale=1"><title>Traffic Squad</title>
<style>
body{font-family:Arial;margin:0;background:#f3f6fa;color:#263746}header,footer{background:#142f49;color:white;padding:18px}main{max-width:1000px;margin:24px auto;padding:0 16px}.box,.card{background:white;padding:18px;border-radius:10px;margin:12px 0;box-shadow:0 2px 9px #0001}button{background:#176f9b;color:white;border:0;padding:10px 14px;border-radius:6px;cursor:pointer}input,select,textarea{display:block;width:100%;padding:10px;margin:7px 0 13px;box-sizing:border-box}label{font-weight:bold}.hidden{display:none}.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(220px,1fr));gap:12px}img{max-width:180px;display:block}nav button{margin:4px}
</style></head><body>
<header><h1> 🚦 Traffic Squad</h1><nav><button onclick="page('home')">Home</button><button onclick="page('register')">Register</button><button onclick="page('login')">Login</button><button onclick="page('cases')">My Challans</button><button onclick="page('police')">Police Desk</button><button onclick="page('admin')">Admin Report</button><button onclick="logout()">Logout</button></nav></header>
<main><div id="msg"></div>
<section id="home"><div class="box"><h2>Traffic Penalty & Case Management</h2><p>Digital traffic cases, photographic proof and challan details.</p><a target="_blank" href="https://www.google.com/maps">Open Google Maps traffic</a></div></section>
```

```
<section id="register" class="hidden box"><h2>User Registration</h2><form
id="reg"><label>Name</label><input id="rn" required><label>Email</label><input id="re"
type="email" required><label>Password</label><input id="rp" type="password"
required><button>Register</button></form></section>
<section id="login" class="hidden box"><h2>Login</h2><form id="log"><label>Email</label><input
id="le" type="email" required><label>Password</label><input id="lp" type="password"
required><button>Login</button></form><p>Demo police: police@trafficsquad.test / police123;
admin: admin@trafficsquad.test / admin123</p></section>
<section id="cases" class="hidden box"><h2>My Challans / Cases</h2><div
id="mycases"></div></section>
<section id="police" class="hidden box"><h2>Create Traffic Case</h2><form
id="caseform"><label>Owner name</label><input id="owner" required><label>Owner
email</label><input id="ownerEmail" type="email" required><label>Vehicle number</label><input
id="vehicle" required><label>Violation rule</label><select id="rule"><option>Signal
jumping</option><option>No helmet</option><option>Over speeding</option><option>No
parking</option></select><label>Fine amount (₹)</label><input id="fine" type="number" min="0"
required><label>Location</label><input id="loc" required><label>Proof photo</label><input
id="proof" type="file" accept="image/*"><button>Create Challan</button></form></section>
<section id="admin" class="hidden box"><h2>Daily Report</h2><div id="report"></div></section>
</main><footer>Traffic Squad – Lab Prototype</footer>
<script>
const
get=k=>JSON.parse(localStorage.getItem(k)||"[]"),set=(k,v)=>localStorage.setItem(k,JSON.stringify(v))
;
let role="",email="";
function say(t){document.getElementById("msg").innerHTML='<p class="box">'+t+'</p>'}
function page(id){document.querySelectorAll("main
section").forEach(s=>s.classList.add("hidden"));document.getElementById(id).classList.remove("hidde
n");if(id==="cases")renderCases();if(id==="admin")renderReport()}
document.getElementById("reg").onsubmit=e=>{e.preventDefault();let
a=get("tsUsers");a.push({name:rn.value,email:re.value.toLowerCase(),password:rp.value});set("tsUsers
",a);say("Registration completed.");page("login")};
document.getElementById("log").onsubmit=e=>{e.preventDefault();email=le.value.toLowerCase();if(e
mail==="police@trafficsquad.test"&&lp.value==="police123")role="police";else
if(email==="admin@trafficsquad.test"&&lp.value==="admin123")role="admin";else
if(get("tsUsers").some(u=>u.email===email&&u.password===lp.value))role="user";else{return
say("Invalid login.")}say("Logged in as
"+role);page(role==="police"?"police":role==="admin"?"admin":"cases")};
document.getElementById("caseform").onsubmit=e=>{e.preventDefault();if(role!=="police")return
say("Police login required.");let f=proof.files[0];let save=()=>{let
a=get("tsCases");a.push({id:"TS"+Date.now(),owner:owner.value,email:ownerEmail.value.toLowerCas
e(),vehicle:vehicle.value.toUpperCase(),rule:rule.value,fine:fine.value,location:loc.value,photo:window.
proofData||"",date:new Date().toLocaleDateString()});set("tsCases",a);say("Challan created
```

```
successfully.");e.target.reset()};if(f){let r=new
FileReader();r.onload=()=>{window.proofData=r.result;save()};r.readAsDataURL(f)}else save()};
function renderCases(){let
a=get("tsCases").filter(c=>c.email===email);document.getElementById("mycases").innerHTML=a.leng
th?a.map(c=>`<article class="card"><b>${c.id}</b><p>${c.vehicle} – ${c.rule}</p><p>Fine:
₹${c.fine} | ${c.date}</p><p>${c.location}</p>${c.photo?`<img
src="${c.photo}">`:""}</article>`).join(""):"No challans found for this login."}
function renderReport(){let a=get("tsCases").filter(c=>c.date===new
Date().toLocaleDateString());document.getElementById("report").innerHTML=`Cases today:
${a.length}<ul>${a.map(c=>`<li>${c.vehicle} – ${c.rule} – ₹${c.fine}</li>`).join("")}</ul>`}
function logout(){role="";email="";say("Logged out.");page("home")}
</script></body></html>
```

8. Sample Input

Operation	Field	Sample value
User registration	Name / Email / Password	Kiran / kiran@example.com / user123
Police login	Email / Password	police@trafficsquad.test / police123
Traffic case	Vehicle / Rule / Fine	KA01AB1234 / Signal jumping / 1000
Evidence	Photo	traffic-proof.jpg

9. Expected Output with Image

The browser displays the project homepage and navigation options. Users can interact with the forms and view the relevant records or confirmation messages. The exact output depends on the sample data entered during testing.

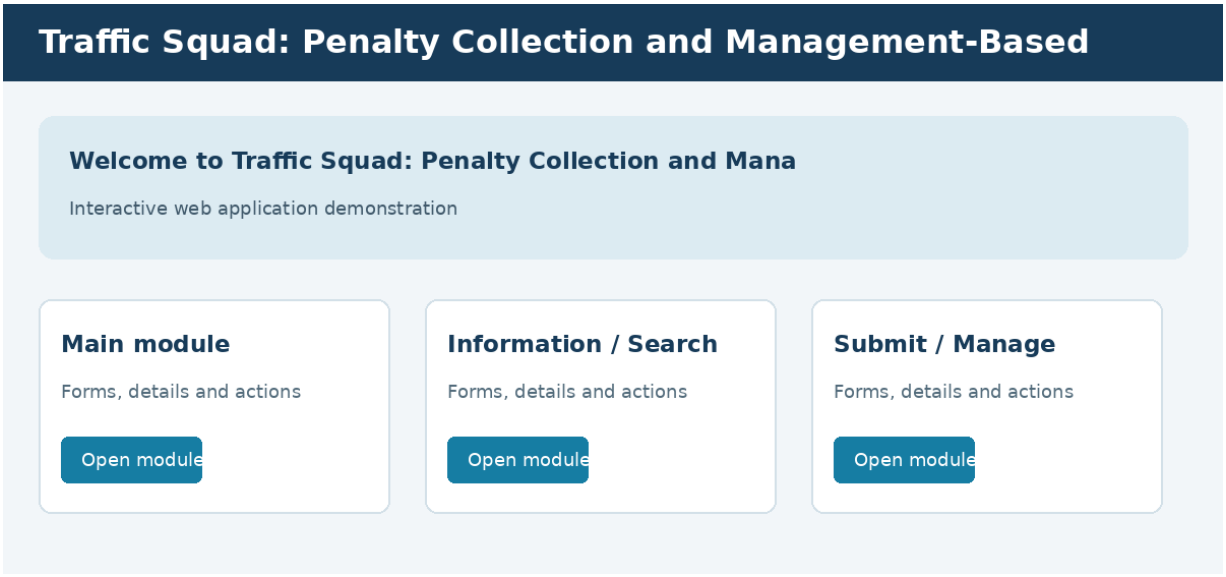


Figure: Illustrative expected output for Experiment 26.

10. Result

Thus, Traffic Squad: Penalty Collection and Management-Based Web Application was designed and developed successfully using HTML, CSS and JavaScript. The main modules and sample workflows were demonstrated in a web browser.

Safety note: this is not an official government challan service. A real deployment requires police authorization, secure evidence storage, audited access and legally compliant payment and appeal processes.

EXPERIMENT 27

1. Title

Online Bus Pass and E-Ticket Management System

2. Aim

To develop a web application for applying for student and route bus passes, renewing passes, adding wallet money, purchasing e-tickets and enabling conductors to verify QR/pass details and update bus location.

3. Software Requirements

Component	Requirement
Frontend	HTML5
Styling	CSS3
Programming	JavaScript (ES6)
Editor	Visual Studio Code / text editor
Browser	Google Chrome / Microsoft Edge / Firefox
Storage	Browser LocalStorage (prototype only)
OS	Windows / Linux / macOS

4. Problem Description

Manual bus pass applications and ticket verification can be slow and difficult to maintain. This system organizes passenger, route, bus, pass and conductor information. Admin manages routes and approvals; users apply for passes and tickets; conductors verify pass/ticket references and update bus location. Government-school student passes can be marked free, while other pass fees are calculated according to the selected validity period.

5. Modules

Admin

- Login
- Manage student and route passes
- Manage conductor login
- Approve bus pass
- Generate pass QR code
- Create bus details
- Create route information: origin, destination, time and stops

User

- Register
- Login
- Add money to e-wallet
- Buy bus e-ticket
- Apply for new bus pass
- Renew bus pass
- Apply student and route pass

- View pass QR code
- Calculate expected arrival time

Conductor

- Login
- Scan or enter QR/pass ID
- Verify bus pass
- Verify bus ticket
- Update bus location

6. Algorithm

9. Initialize demo route, bus, user, pass and ticket data.
10. User registers and logs in, then adds wallet balance if required.
11. User selects a route and applies for a student or route pass with required details and validity.
12. Calculate the pass amount; government-school student pass is free in this prototype.
13. Admin reviews and approves pass applications; approved passes receive a generated reference/QR representation.
14. User may buy a ticket, with fare deducted from available wallet balance.
15. Conductor enters or scans the pass/ticket reference and verifies it against stored records.
16. Conductor updates bus location; user views route and estimated travel information.
17. End.

7. Complete HTML, CSS and JavaScript Code

```
<!DOCTYPE html><html lang="en"><head><meta charset="UTF-8"><meta name="viewport"
content="width=device-width,initial-scale=1"><title>Bus Pass Portal</title><style>
body{font-
family:Arial;margin:0;background:#f4f7fb;color:#243746}header,footer{background:#173b59;color:wh
ite;padding:18px}main{max-width:1000px;margin:22px auto;padding:0
16px}.box,.card{background:white;padding:18px;border-radius:10px;margin:12px
0}button{background:#167da3;color:white;border:0;padding:10px;border-
radius:6px;cursor:pointer;margin:3px}input,select{width:100%;padding:10px;margin:7px 0 12px;box-
sizing:border-box}.hidden{display:none}nav{display:flex;flex-wrap:wrap}label{font-weight:bold}
</style></head><body><header><h1>🚌 Smart Bus Pass Portal</h1><nav><button
onclick="show('home')">Home</button><button onclick="show('register')">Register</button><button
onclick="show('login')">Login</button><button onclick="show('apply')">Apply Pass</button><button
onclick="show('wallet')">Wallet</button><button
onclick="show('conductor')">Conductor</button><button
onclick="show('admin')">Admin</button></nav></header><main><div id="msg"></div>
<section id="home"><div class="box"><h2>Bus Pass & E-Ticket Services</h2><p>Apply, renew and
verify digital bus passes.</p><div id="routeInfo"></div></div></section>
<section id="register" class="box hidden"><h2>Register</h2><form
id="reg"><label>Name</label><input id="name" required><label>Email</label><input id="email"
type="email" required><label>Password</label><input id="password" type="password"
required><button>Register</button></form></section>
```

```
<section id="login" class="box hidden"><h2>Login</h2><form
id="login"><label>Email</label><input id="liEmail" type="email"
required><label>Password</label><input id="liPass" type="password"
required><button>Login</button></form><p>Demo admin: admin@bus.test / admin123; conductor:
conductor@bus.test / cond123</p></section>
<section id="wallet" class="box hidden"><h2>eWallet</h2><p id="balance"></p><input id="amount"
type="number" min="1" placeholder="Amount to add"><button onclick="addMoney()">Add
Money</button><button onclick="buyTicket()">Buy sample ticket (₹25)</button></section>
<section id="apply" class="box hidden"><h2>Apply / Renew Bus Pass</h2><form
id="passForm"><label>Pass type</label><select
id="type"><option>Student</option><option>Route</option></select><label>School
type</label><select id="school"><option>Government</option><option>Private</option><option>Not
applicable</option></select><label>Route</label><select id="route"><option>Town to
College</option><option>Town to Market</option></select><label>Validity</label><select
id="months"><option value="1">1 month</option><option value="3">3
months</option></select><label>Organization / School</label><input id="org"
required><button>Submit Pass Application</button></form><div id="myPass"></div></section>
<section id="conductor" class="box hidden"><h2>Conductor verification</h2><input id="passid"
placeholder="Enter pass or ticket ID"><button onclick="verify()">Verify</button><p
id="verifyResult"></p><label>Update bus location</label><input id="busloc" placeholder="Current
stop"><button onclick="updateLocation()">Update Location</button></section>
<section id="admin" class="box hidden"><h2>Admin approval and route setup</h2><div
id="pending"></div><form id="routeForm"><label>From</label><input id="from"
required><label>To</label><input id="to" required><label>Time</label><input id="time"
required><label>Stops</label><input id="stops" required><button>Add
Route</button></form></section>
</main><footer>Smart Bus Pass – Lab Prototype</footer><script>
const
read=k=>JSON.parse(localStorage.getItem("bus_"+k)||"null"),write=(k,v)=>localStorage.setItem("bus_"
+k,JSON.stringify(v));
if(!read("passes"))write("passes",[]);if(!read("users"))write("users",[]);if(!read("routes"))write("routes",[
{from:"Town",to:"College",time:"08:00 AM",stops:"Main Road, Market"}]);let who="",role="";
function say(t){document.getElementById("msg").textContent=t}
function show(id){document.querySelectorAll("main
section").forEach(x=>x.classList.add("hidden"));document.getElementById(id).classList.remove("hidde
n");if(id==="wallet")document.getElementById("balance").textContent="Wallet balance:
₹"+(read("wallet_"+who)||0);if(id==="admin")renderAdmin();if(id==="home")document.getElementBy
Id("routeInfo").textContent=read("routes").map(r=>r.from+" → "+r.to+" | "+r.time+" | Stops:
"+r.stops).join(" ; ")}
document.getElementById("reg").onsubmit=e=>{e.preventDefault();let
a=read("users");a.push({name:name.value,email:email.value,password:password.value});write("users",a
);say("Registered. Please log in.");show("login")};
document.getElementById("login").onsubmit=e=>{e.preventDefault();who=liEmail.value;let
u=read("users").find(x=>x.email===who&&x.password===liPass.value);if(who==="admin@bus.test"
```

```
&&liPass.value==="admin123")role="admin";else
if(who==="conductor@bus.test"&&liPass.value==="cond123")role="conductor";else
if(u)role="user";else return say("Invalid login");say("Logged in as
"+role);show(role==="admin"?"admin":role==="conductor"?"conductor":"wallet")};
function addMoney(){let
n=Number(amount.value);if(n>0){write("wallet_"+who,(read("wallet_"+who)||0)+n);show("wallet");say
("Wallet updated.")}}
function buyTicket(){let b=read("wallet_"+who)||0;if(b<25)return say("Insufficient wallet
balance.");write("wallet_"+who,b-25);let id="TKT"+Date.now();let
t=read("tickets")||[];t.push({id,owner:who,fare:25});write("tickets",t);say("Ticket purchased. Ticket ID:
"+id);show("wallet")}
document.getElementById("passForm").onsubmit=e=>{e.preventDefault();if(!who)return say("Login
first");let free=type.value==="Student"&&school.value==="Government";let
cost=free?0:Number(months.value)*500;let a=read("passes");let
p={id:"PASS"+Date.now(),email:who,type:type.value,school:school.value,route:route.value,months:mo
nths.value,org:org.value,cost,status:"Pending"};a.push(p);write("passes",a);document.getElementById("
myPass").textContent="Application "+p.id+" submitted. Amount: ₹"+cost+"; status:
Pending.";say("Pass application submitted.")};
function
renderAdmin(){document.getElementById("pending").innerHTML=read("passes").map(p=>`<p>${p.id
} – ${p.email} – ₹${p.cost} – ${p.status} <button
onclick="approve('${p.id}')">Approve</button></p>`).join("")||"No applications."}
function approve(id){let a=read("passes");let
p=a.find(x=>x.id===id);if(p)p.status="Approved";write("passes",a);renderAdmin()}
function verify(){let id=passid.value;let
p=read("passes").find(x=>x.id===id&&x.status==="Approved");let
t=(read("tickets")||[]).find(x=>x.id===id);document.getElementById("verifyResult").textContent=p?"Va
lid approved bus pass":t?"Valid e-ticket":"Pass or ticket not found / not approved"}
function updateLocation(){write("bus_location",busloc.value);say("Bus location updated:
"+busloc.value)}
document.getElementById("routeForm").onsubmit=e=>{e.preventDefault();let
a=read("routes");a.push({from:from.value,to:to.value,time:time.value,stops:stops.value});write("routes",
a);e.target.reset();renderAdmin();say("Route added.")};
show("home");
</script></body></html>
```

8. Sample Input

Operation	Field	Sample value
User	Name / Email	Meena / meena@example.com
Route	From / To / Stops	Town / College / Main Road, Market
Student pass	School type / Validity	Government / 1 month
Ticket	Route / Fare	Town–College / ₹25
Conductor verification	Pass ID	PASS-1001

9. Expected Output with Image

The browser displays the project homepage and navigation options. Users can interact with the forms and view the relevant records or confirmation messages. The exact output depends on the sample data entered during testing.

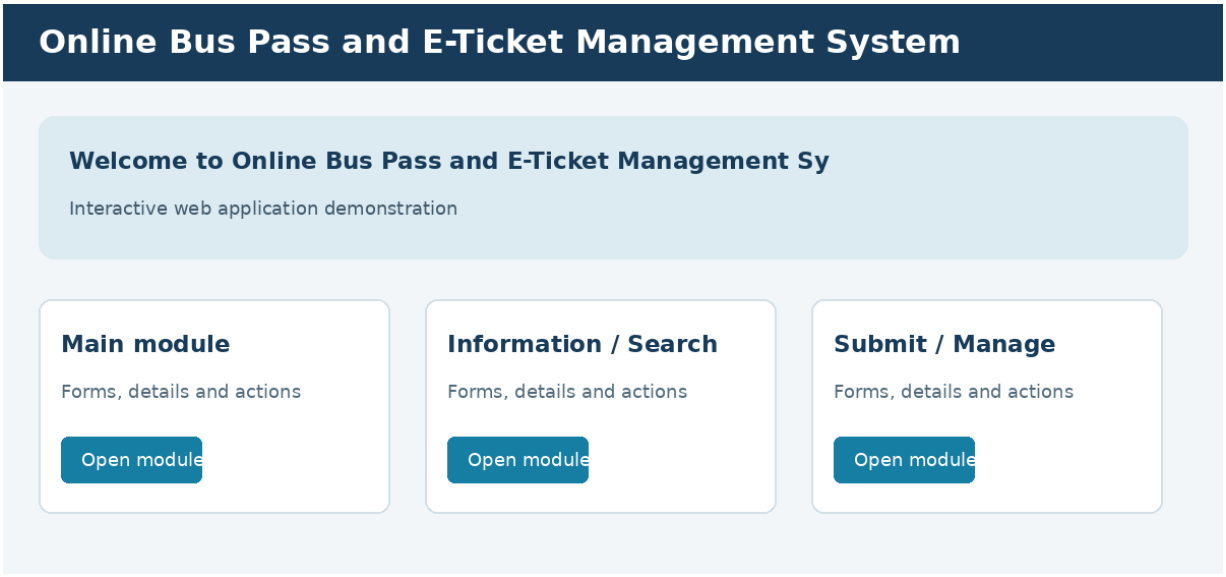


Figure: Illustrative expected output for Experiment 27.

10. Result

Thus, Online Bus Pass and E-Ticket Management System was designed and developed successfully using HTML, CSS and JavaScript. The main modules and sample workflows were demonstrated in a web browser.

QR note: this prototype uses a pass reference that can be entered for verification. A production QR workflow should generate and validate signed QR tokens using a backend.

EXPERIMENT 28A

1. Title
ArtistryHub – Artist Portfolio Website

2. Aim
To design an attractive artist portfolio website that presents an artist biography, artwork gallery, exhibitions, media, news and contact information in a responsive layout.

3. Software Requirements

Component	Requirement
Frontend	HTML5
Styling	CSS3
Programming	JavaScript (ES6)
Editor	Visual Studio Code / text editor
Browser	Google Chrome / Microsoft Edge / Firefox
Storage	Browser LocalStorage (prototype only)
OS	Windows / Linux / macOS

4. Problem Description
Artists need a digital space to introduce their background and present their creative work to visitors. ArtistryHub provides a portfolio with an engaging homepage, biography, artwork gallery, selected works, news and contact form. The sample artist Emily Harris and sample artwork descriptions are fictional demonstration content and can be replaced with authorized material.

5. Modules

Visitor
- View homepage and featured artwork
- Read artist biography
- Browse portfolio gallery
- View exhibitions and news
- Send contact / booking inquiry

Artist / Site Editor (prototype)
- Update artwork entries in the JavaScript data
- Update biography and exhibition details in page content

6. Algorithm
18. Load the portfolio homepage and featured artwork.
19. Display artist biography and career highlights.
20. Read the artwork data array and create gallery cards.
21. Display exhibition and news items.
22. Allow a visitor to enter a contact message and validate required fields.
23. Show a confirmation message for the sample form submission.
24. End.

7. Complete HTML, CSS and JavaScript Code

```
<!DOCTYPE html><html lang="en"><head><meta charset="UTF-8"><meta name="viewport"
content="width=device-width,initial-scale=1"><title>ArtistryHub</title><style>
*{box-sizing:border-box}body{margin:0;font-
family:Georgia,serif;background:#faf7f2;color:#302b2a}header{background:#292331;color:white;paddi
ng:22px 5%;display:flex;justify-content:space-between;flex-wrap:wrap}nav
a{color:white;margin:8px;text-decoration:none}.hero{padding:70px 8%;background:linear-
gradient(120deg,#e9d8ca,#c6b5c9)}.hero h1{font-size:48px;margin:0}.wrap{max-
width:1100px;margin:auto;padding:35px 20px}.gallery{display:grid;grid-template-columns:repeat(auto-
fit,minmax(220px,1fr));gap:18px}.art{background:white;padding:12px;border-radius:9px}.art
img{width:100%;height:190px;object-fit:cover;border-radius:6px}.swatch{height:190px;border-
radius:6px;background:linear-gradient(135deg,#8ba7a8,#d8a98e,#665b78)}.art:nth-child(2)
.swatch{background:linear-gradient(135deg,#d5b16e,#a65c50,#453d56)}.art:nth-child(3)
.swatch{background:linear-gradient(135deg,#8da9c4,#e4d6c2,#607d72)}section{scroll-margin-
top:15px}input,textarea{width:100%;padding:11px;margin:7px 0
13px}button{background:#59445d;color:white;padding:11px
18px;border:0;cursor:pointer}footer{background:#292331;color:white;text-align:center;padding:20px}
</style></head><body><header><h2>ArtistryHub</h2><nav><a href="#about">About</a><a
href="#portfolio">Portfolio</a><a href="#exhibitions">Exhibitions</a><a href="#news">News</a><a
href="#contact">Contact</a></nav></header>
<div class="hero"><h1>Emily Harris</h1><p>Art, imagination and stories in every brushstroke.</p><a
href="#portfolio">Explore the Portfolio ↓</a></div>
<section id="about" class="wrap"><h2>About the Artist</h2><p>Emily Harris is a fictional
contemporary visual artist whose sample portfolio explores nature, memory and colour. Inspired by
landscapes, everyday life and changing light, her work combines expressive textures with thoughtful
composition. Replace this sample biography with the artist's verified biography and portrait before
publishing.</p></section>
<section id="portfolio" class="wrap"><h2>Selected Works</h2><div id="gallery"
class="gallery"></div></section>
<section id="exhibitions" class="wrap"><h2>Exhibitions & Works</h2><ul><li>Light & Form –
December 2026, Sample Gallery</li><li>Quiet Landscapes – August 2026, Community Arts
Hall</li><li>Colour Stories – March 2026, Online showcase</li></ul></section>
<section id="news" class="wrap"><h2>News & Updates</h2><article class="art"><h3>Studio
Notes</h3><p>New works and exhibition announcements will appear here.</p></article></section>
<section id="contact" class="wrap"><h2>Contact & Booking</h2><form
id="contactForm"><label>Name</label><input id="visitor" required><label>Email</label><input
id="mail" type="email" required><label>Message</label><textarea id="message"
required></textarea><button>Send Inquiry</button></form><p id="response"></p></section>
<footer>© 2026 ArtistryHub – Artist Portfolio Demonstration</footer>
<script>
const works=[{title:"Blue Horizon",medium:"Acrylic on canvas",year:"2026",desc:"A study of distance
and calm."},{title:"Golden Hour",medium:"Mixed media",year:"2025",desc:"Warm colours inspired by
evening light."},{title:"Quiet Garden",medium:"Watercolour",year:"2025",desc:"A delicate view of a
peaceful garden."}];
```

```
document.getElementById("gallery").innerHTML=works.map((w,i)=>`<article class="art"><div
class="swatch" role="img" aria-label="Decorative colour artwork
placeholder"></div><h3>${w.title}</h3><p>${w.medium} ·
${w.year}</p><p>${w.desc}</p></article>`).join("");
document.getElementById("contactForm").onsubmit=e=>{e.preventDefault();document.getElementById("response").textContent="Thank you, "+visitor.value+". Your sample inquiry has been recorded on this page.";e.target.reset()};
</script></body></html>
```

8. Sample Input

Operation	Field	Sample value
Artwork	Title / Medium	Blue Horizon / Acrylic on canvas
Exhibition	Name / Date	Light & Form / 15 December 2026
Contact	Name / Email / Message	Asha / asha@example.com / Booking inquiry

9. Expected Output with Image

The browser displays the project homepage and navigation options. Users can interact with the forms and view the relevant records or confirmation messages. The exact output depends on the sample data entered during testing.

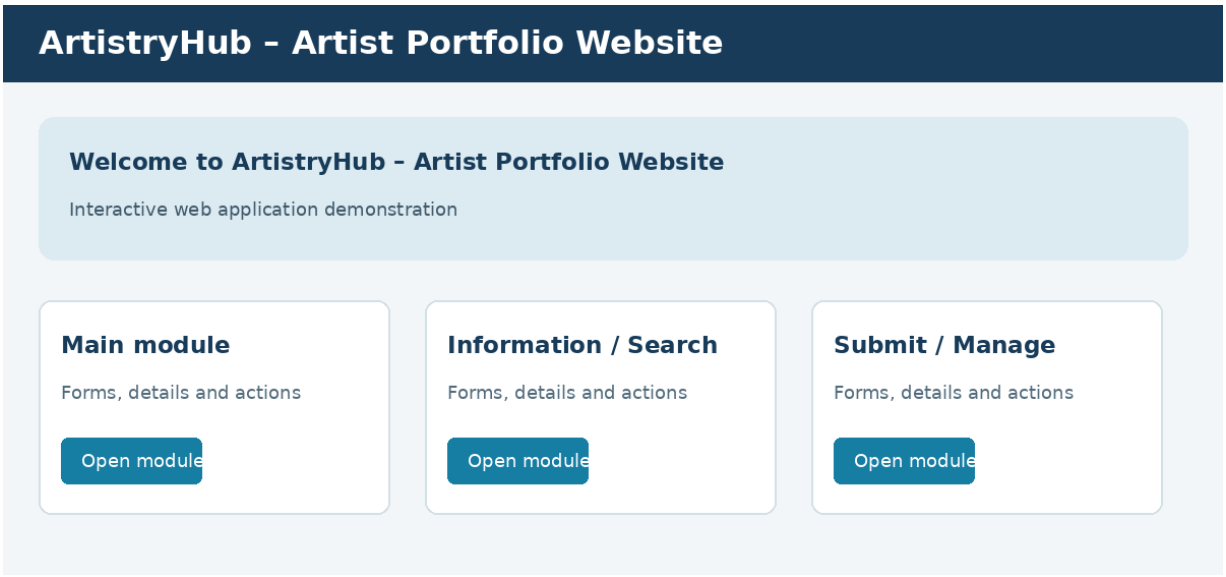


Figure: Illustrative expected output for Experiment 28A.

10. Result

Thus, ArtistryHub – Artist Portfolio Website was designed and developed successfully using HTML, CSS and JavaScript. The main modules and sample workflows were demonstrated in a web browser.

EXPERIMENT 29

1. Title
Event Information and Online Registration Website

2. Aim
To create a responsive event webpage that presents event purpose, target audience, speakers, venue, schedule and an online registration form.

3. Software Requirements

Component	Requirement
Frontend	HTML5
Styling	CSS3
Programming	JavaScript (ES6)
Editor	Visual Studio Code / text editor
Browser	Google Chrome / Microsoft Edge / Firefox
Storage	Browser LocalStorage (prototype only)
OS	Windows / Linux / macOS

4. Problem Description
Event information can be scattered across multiple announcements, making it difficult for interested participants to understand the event and register. This website organizes the event overview, speaker information, venue, agenda and registration form into clear sections.

5. Modules

Visitor
- View event homepage and purpose
- View speakers and speaker links
- View venue and schedule
- Read target audience information
- Register for the event

Organizer (content maintenance)
- Edit event details, speaker names, venue and schedule in the page source

6. Algorithm
25. Load the event landing page.
26. Display the event title, date, venue and purpose.
27. Show speaker cards with links and the event agenda.
28. Explain the intended audience and event outcomes.
29. Collect attendee name, email and ticket type in the registration form.
30. Validate the form and display a registration acknowledgement.
31. End.

7. Complete HTML, CSS and JavaScript Code

```
<!DOCTYPE html><html lang="en"><head><meta charset="UTF-8"><meta name="viewport"
content="width=device-width,initial-scale=1"><title>Future Forward Summit</title><style>
body{margin:0;font-
family:Arial;background:#f4f6fb;color:#24364b}header{background:#172b4d;color:white;padding:18p
x 6%;display:flex;justify-content:space-between;flex-wrap:wrap}nav a{color:white;margin:8px;text-
decoration:none}.hero{text-align:center;padding:65px 18px;background:linear-
gradient(120deg,#dce9ff,#f4e7d8)}.hero h1{font-
size:42px;color:#172b4d}.btn,button{background:#e36c45;color:white;border:0;border-
radius:6px;padding:12px 18px;cursor:pointer;text-decoration:none}.wrap{max-
width:1050px;margin:25px auto;padding:0 18px}.panel{background:white;padding:20px;border-
radius:10px;margin:15px 0}.grid{display:grid;grid-template-columns:repeat(auto-
fit,minmax(210px,1fr));gap:15px}.speaker{background:white;padding:18px;border-radius:10px;text-
align:center}.avatar{width:90px;height:90px;border-
radius:50%;background:#c8d7ee;margin:auto;display:grid;place-items:center;font-
size:30px}input,select{width:100%;padding:11px;margin:7px 0 13px;box-sizing:border-
box}footer{background:#172b4d;color:white;text-align:center;padding:20px}
</style></head><body><header><b>Future Forward Summit</b><nav><a
href="#about">About</a><a href="#speakers">Speakers</a><a href="#schedule">Schedule</a><a
href="#register">Register</a></nav></header>
<div class="hero"><h1>Future Forward Summit 2026</h1><p>Ideas, innovation and inspiration for
tomorrow.</p><p>20 December 2026 · City Convention Hall</p><a class="btn"
href="#register">Register Now</a></div>
<section id="about" class="wrap panel"><h2>About the Event</h2><p>A one-day learning and
networking event for students, educators, entrepreneurs and technology professionals. The summit
explores digital innovation, career readiness and responsible technology.</p><h3>Who should
attend?</h3><p>College students, early-career professionals, educators, founders and anyone interested
in technology and innovation.</p></section>
<section id="speakers" class="wrap"><h2>Featured Speakers</h2><div class="grid"><article
class="speaker"><div class="avatar">👩</div><h3>Dr. Maya Rao</h3><p>Research and
Innovation</p><a href="#schedule">Session details</a></article><article class="speaker"><div
class="avatar">👨</div><h3>Arjun Mehta</h3><p>Entrepreneurship</p><a
href="#schedule">Session details</a></article><article class="speaker"><div
class="avatar">👩</div><h3>Sana Joseph</h3><p>Responsible Technology</p><a
href="#schedule">Session details</a></article></div></section>
<section id="schedule" class="wrap panel"><h2>Event Schedule</h2><p>09:00 – Registration and
welcome</p><p>10:00 – Keynote: Innovation for the Future</p><p>11:30 – Panel
discussion</p><p>13:00 – Lunch and networking</p><p>14:00 – Workshops and closing
session</p><h3>Venue</h3><p>City Convention Hall, Main Road. <a target="_blank"
href="https://www.google.com/maps">View map</a></p></section>
<section id="register" class="wrap panel"><h2>Participant Registration</h2><form
id="form"><label>Full name</label><input id="name" required><label>Email</label><input
id="email" type="email" required><label>Category</label><select
```

```
id="category"><option>Student</option><option>Professional</option><option>Educator</option><o
ption>Other</option></select><button>Submit Registration</button></form><p
id="result"></p></section>
<footer>© 2026 Future Forward Summit</footer><script>
document.getElementById("form").onsubmit=e=>{e.preventDefault();let
ref="EVT"+Date.now().toString().slice(-
7);document.getElementById("result").textContent="Registration received for "+name.value+".
Reference: "+ref+". This is a front-end demonstration.";e.target.reset()};
</script></body></html>
```

8. Sample Input

Operation	Field	Sample value
Attendee	Name / Email	Arun / arun@example.com
Ticket	Type	Student
Event	Venue / Date	City Convention Hall / 20 December 2026

9. Expected Output with Image

The browser displays the project homepage and navigation options. Users can interact with the forms and view the relevant records or confirmation messages. The exact output depends on the sample data entered during testing.

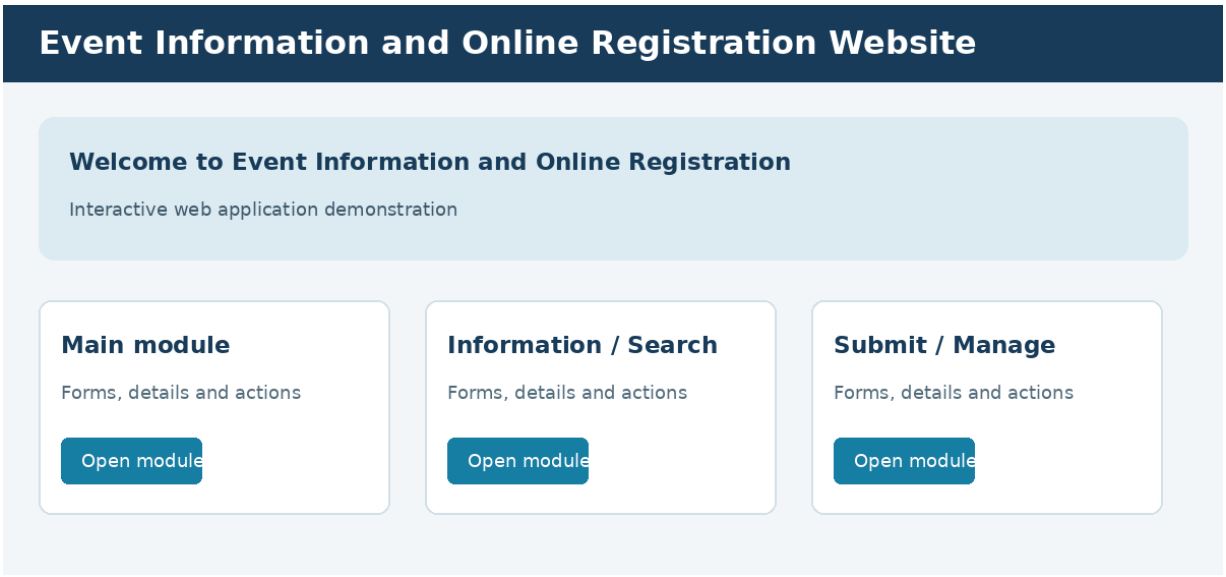


Figure: Illustrative expected output for Experiment 29.

10. Result

Thus, Event Information and Online Registration Website was designed and developed successfully using HTML, CSS and JavaScript. The main modules and sample workflows were demonstrated in a web browser.

EXPERIMENT 30

1. Title

International Journal of Scientific and Social Excellence (IJSSE) – Journal Management Website

2. Aim

To design and develop a journal website that presents journal information, publication ethics, editorial board details, reviewer and author registration, manuscript submission information, article listings and contact details.

3. Software Requirements

Component	Requirement
Frontend	HTML5
Styling	CSS3
Programming	JavaScript (ES6)
Editor	Visual Studio Code / text editor
Browser	Google Chrome / Microsoft Edge / Firefox
Storage	Browser LocalStorage (prototype only)
OS	Windows / Linux / macOS

4. Problem Description

Academic journal workflows involve authors, reviewers and editors exchanging manuscripts and decisions. A centralized portal can make journal information and submission steps easier to access. This lab prototype provides role-based registration/login demonstrations and a manuscript submission form with a local status list. It does not perform actual peer review or verify indexing claims.

5. Modules

Public

- Home page
- About Journal
- Publication Ethics
- Editorial Board
- Articles: current issue, archive and sample most-viewed/cited lists
- Call for Papers
- Indexed-list information (admin-maintained)
- Newsletter subscription
- Contact Us

Editor

- Editor registration
- Editor login
- Editorial committee
- Editor responsibilities

Reviewer

- Reviewer registration

- Reviewer login
- View sample assigned article list

Author

- Author registration
- Author login
- List papers under author
- Author instructions and guidelines
- Publication charge information
- Article template and copyright form template links/placeholders
- Submit manuscript and view status

6. Algorithm

32. Display journal homepage and navigation.
33. Allow authors, reviewers and editors to register with a selected role.
34. Validate login using locally stored demonstration accounts.
35. Author enters manuscript title, abstract and file name and submits the record.
36. Store manuscript with Submitted status and show it in the author's paper list.
37. Display sample reviewer and editorial information pages.
38. Show article lists and journal policy information.
39. End.

7. Complete HTML, CSS and JavaScript Code

```
<!DOCTYPE html><html lang="en"><head><meta charset="UTF-8"><meta name="viewport"
content="width=device-width,initial-scale=1"><title>IJSSE Journal Portal</title><style>
body{font-
family:Arial;margin:0;background:#f5f7fa;color:#293746}header{background:#193b59;color:white;pad
ding:18px 5%}nav{display:flex;flex-wrap:wrap;gap:5px}nav
button,button{background:#176e96;color:white;border:0;padding:9px 12px;border-
radius:5px;cursor:pointer}.wrap{max-width:1100px;margin:22px auto;padding:0
16px}.panel{background:white;padding:18px;border:1px solid #dce5ec;border-radius:9px;margin:12px
0}.hidden{display:none}input,textarea,select{width:100%;padding:10px;margin:6px 0 12px;box-
sizing:border-box}footer{background:#193b59;color:white;text-align:center;padding:18px}
</style></head><body><header><h1>International Journal of Scientific and Social
Excellence</h1><p>Print ISSN: 2395-1443 | Online ISSN: 2395-1451</p><nav><button
onclick="show('home')">Home</button><button onclick="show('about')">About</button><button
onclick="show('ethics')">Ethics</button><button onclick="show('articles')">Articles</button><button
onclick="show('register')">Register</button><button onclick="show('login')">Login</button><button
onclick="show('submit')">Submit Paper</button><button onclick="show('papers')">My
Papers</button><button onclick="show('contact')">Contact</button></nav></header><main
class="wrap"><div id="msg"></div>
<section id="home" class="panel"><h2>Welcome to IJSSE</h2><p>A sample academic journal portal
supporting authors, reviewers and editors.</p><p>Explore journal scope, publication ethics, article
information and submission guidance.</p></section>
```

```
<section id="about" class="panel hidden"><h2>About Journal</h2><p>IJSSE publishes scholarly work in scientific and social disciplines. Add the journal's verified mission, scope, publisher details and indexing status before public deployment.</p><h3>Editorial Board</h3><p>Editor-in-Chief: Sample Editor · Editorial Committee: Subject-area editors.</p><h3>Editor responsibilities</h3><p>Coordinate peer review, maintain confidentiality, manage conflicts of interest and communicate editorial decisions.</p></section>
<section id="ethics" class="panel hidden"><h2>Publication Ethics</h2><p>Authors must submit original work, cite sources accurately and disclose conflicts of interest. Reviewers should provide fair, confidential and timely assessments. Editors should make impartial decisions and handle corrections responsibly.</p></section>
<section id="articles" class="panel hidden"><h2>Articles</h2><h3>Current issue</h3><p>Sample: Digital Inclusion and Community Development.</p><h3>Archive</h3><p>Volume 1 – Sample archive entry.</p><h3>Most viewed / cited</h3><p>Sample article metrics for demonstration only.</p></section>
<section id="register" class="panel hidden"><h2>Registration</h2><form id="reg"><label>Name</label><input id="rname" required><label>Email</label><input id="remail" type="email" required><label>Password</label><input id="rpass" type="password" required><label>Role</label><select id="role"><option>Author</option><option>Reviewer</option><option>Editor</option></select><button>Register</button></form></section>
<section id="login" class="panel hidden"><h2>Login</h2><form id="log"><label>Email</label><input id="lemail" type="email" required><label>Password</label><input id="lpass" type="password" required><button>Login</button></form></section>
<section id="submit" class="panel hidden"><h2>Submit Manuscript</h2><form id="paper"><label>Paper title</label><input id="title" required><label>Abstract</label><textarea id="abstract" required></textarea><label>Manuscript filename (demo)</label><input id="filename" placeholder="paper.docx" required><button>Submit Manuscript</button></form></section>
<section id="papers" class="panel hidden"><h2>My Papers / Assigned Articles</h2><div id="paperList"></div></section>
<section id="contact" class="panel hidden"><h2>Contact Us</h2><p>Email: editor@example.org (sample contact – replace with official address).</p><h3>Author resources</h3><p>Author instructions, publication charge schedule, article template and copyright form should be supplied as approved journal documents.</p><p>Newsletter: <input id="newsletter" type="email" placeholder="Your email"><button onclick="subscribe()">Subscribe</button></p></section>
</main><footer>IJSSE Journal Portal – Academic Lab Prototype</footer><script>
const get=k=>JSON.parse(localStorage.getItem("ijsse_"+k)||"null"),set=(k,v)=>localStorage.setItem("ijsse_"+k,JSON.stringify(v));if(!get("users"))set("users",[]);if(!get("papers"))set("papers",[]);let user=null;
function msg(t){document.getElementById("msg").textContent=t}
function show(id){document.querySelectorAll("main section").forEach(s=>s.classList.add("hidden"));document.getElementById(id).classList.remove("hidden");if(id==="papers")renderPapers()}
```

```
document.getElementById("reg").onsubmit=e=>{e.preventDefault();let
a=get("users");a.push({name:rname.value,email:remail.value,password:rpass.value,role:role.value});set(
"users",a);msg("Registration completed.");e.target.reset();show("login")};
document.getElementById("log").onsubmit=e=>{e.preventDefault();user=get("users").find(u=>u.email
===lemail.value&&u.password===lpass.value);if(!user)return msg("Invalid login
details.");msg("Welcome "+user.name+"
("+user.role+").");show(user.role==="Author"?"submit":"papers")};
document.getElementById("paper").onsubmit=e=>{e.preventDefault();if(!user||user.role!=="Author")ret
urn msg("Please log in as an author.");let
a=get("papers");a.push({id:"IJSSE"+Date.now(),email:user.email,title:title.value,abstract:abstract.value,
file:filename.value,status:"Submitted"});set("papers",a);msg("Manuscript submitted
successfully.");e.target.reset();renderPapers()};
function renderPapers(){let
a=get("papers").filter(p=>user&&p.email===user.email);document.getElementById("paperList").inner
HTML=a.map(p=>`<p>${p.id} – ${p.title} – Status: ${p.status}</p>`).join("")||"No papers found for
this account."}
function subscribe(){msg("Newsletter subscription captured for this demonstration.")}
</script></body></html>
```

8. Sample Input

Operation	Field	Sample value
Author registration	Name / Email / Role	Priya / priya@example.com / Author
Manuscript	Title / Abstract	Digital Inclusion / Study of access to digital services
Reviewer	Name / Email	Dr. Rao / reviewer@example.com
Editor	Email / Password	editor@example.com / editor123

9. Expected Output with Image

The browser displays the project homepage and navigation options. Users can interact with the forms and view the relevant records or confirmation messages. The exact output depends on the sample data entered during testing.



Figure: Illustrative expected output for Experiment 30.

10. Result

Thus, International Journal of Scientific and Social Excellence (IJSSE) – Journal Management Website was designed and developed successfully using HTML, CSS and JavaScript. The main modules and sample workflows were demonstrated in a web browser.

Journal note: ISSN values are included as supplied in the project brief. Any Scopus-indexing claim, journal policy, publication charge, editorial identity and template must be verified by the journal before publication.

EXPERIMENT 31

1. Title

Ko-Matha Dairy Farm – Dairy Products and Farm Management Website

2. Aim

To develop a dairy farm website that presents milk and dairy products, supports sample online orders and maintains administrator, stock, customer, vendor and cow information with daily, monthly and yearly stock reports.

3. Software Requirements

Component	Requirement
Frontend	HTML5
Styling	CSS3
Programming	JavaScript (ES6)
Editor	Visual Studio Code / text editor
Browser	Google Chrome / Microsoft Edge / Firefox
Storage	Browser LocalStorage (prototype only)
OS	Windows / Linux / macOS

4. Problem Description

Dairy businesses need to present products to customers while tracking stock, orders, suppliers and farm assets. Ko-Matha Dairy Farm's website combines public product information and a demonstration order form with basic management forms for stock, customer purchases, milk vendors and cow details.

5. Modules

Public

- Home
- About Us
- Products
- Order Online
- Contact

Administrator

- Admin registration and login
- Stock entry
- Daily report
- Monthly report
- Yearly report

Customer

- Customer details
- Purchase details
- Online purchase order

Milk Vendor / Farm

- Vendor information
- Cow detail maintenance

6. Algorithm

40. Load sample dairy products and initialize stock, orders, vendor and cow records.
41. Display homepage, farm information and products with prices.
42. Admin registers or logs in and enters stock quantities.
43. Customer submits an online product order with quantity and contact details.
44. Validate stock availability and reduce available stock for a successful sample order.
45. Store purchase details and show order acknowledgement.
46. Admin views daily, monthly and yearly order/stock summaries.
47. Admin adds or reviews vendor and cow information.
48. End.

7. Complete HTML, CSS and JavaScript Code

```
<!DOCTYPE html><html lang="en"><head><meta charset="UTF-8"><meta name="viewport"
content="width=device-width,initial-scale=1"><title>Ko-Matha Dairy Farm</title><style>
body{font-
family:Arial;margin:0;background:#f7f8f1;color:#344332}header,footer{background:#365b3d;color:wh
ite;padding:18px}nav{display:flex;flex-
wrap:wrap;gap:5px}button{background:#668b4e;color:white;border:0;padding:10px;border-
radius:5px;cursor:pointer}.wrap{max-width:1050px;margin:20px auto;padding:0
16px}.panel,.card{background:white;padding:18px;border-radius:10px;margin:12px
0}.grid{display:grid;grid-template-columns:repeat(auto-
fit,minmax(200px,1fr));gap:12px}.hidden{display:none}input,select{width:100%;padding:10px;margin:
7px 0 12px;box-sizing:border-box}label{font-weight:bold}
</style></head><body><header><h1>🐄 Ko-Matha Dairy Farm</h1><nav><button
onclick="show('home')">Home</button><button onclick="show('about')">About</button><button
onclick="show('products')">Products</button><button onclick="show('order')">Order
Online</button><button onclick="show('admin')">Admin</button><button
onclick="show('farm')">Farm Records</button></nav></header><main class="wrap"><div
id="msg"></div>
<section id="home" class="panel"><h2>Freshness from our farm</h2><p>Fresh dairy products and
farm information in one place.</p><div id="featured" class="grid"></div></section>
<section id="about" class="panel hidden"><h2>About Ko-Matha</h2><p>Ko-Matha Dairy Farm is a
sample dairy business website concept focused on product quality, farm transparency and customer
convenience. Product and A1/A2 claims should be verified and supported by appropriate testing and
documentation before publication.</p></section>
<section id="products" class="panel hidden"><h2>Products</h2><div id="productList"
class="grid"></div></section>
<section id="order" class="panel hidden"><h2>Online Purchase Order</h2><form
id="orderForm"><label>Customer name</label><input id="customer"
required><label>Phone</label><input id="phone" required><label>Product</label><select
```

```
id="product"></select><label>Quantity</label><input id="qty" type="number" min="1" value="1"
required><button>Place Order</button></form><p id="orderResult"></p></section>
<section id="admin" class="panel hidden"><h2>Administrator Login</h2><form
id="adminLogin"><label>Email</label><input id="adminEmail" type="email"
required><label>Password</label><input id="adminPass" type="password"
required><button>Login</button></form><div id="adminArea" class="hidden"><h3>Stock
Entry</h3><form id="stockForm"><label>Product</label><select
id="stockProduct"></select><label>Quantity to add</label><input id="stockQty" type="number"
min="1" required><button>Update Stock</button></form><button onclick="report('day')">Daily
Report</button><button onclick="report('month')">Monthly Report</button><button
onclick="report('year')">Yearly Report</button><div id="report"></div></div></section>
<section id="farm" class="panel hidden"><h2>Vendor and Cow Details</h2><form
id="vendorForm"><h3>Add Milk Vendor</h3><label>Vendor name</label><input id="vendor"
required><label>Phone</label><input id="vendorPhone" required><button>Add
Vendor</button></form><form id="cowForm"><h3>Add Cow Record</h3><label>Tag
number</label><input id="tag" required><label>Breed</label><input id="breed"
required><button>Add Cow</button></form><div id="farmRecords"></div></section>
</main><footer>Ko-Matha Dairy Farm – Lab Prototype</footer><script>
const products=[{name:"Fresh Milk (litre)",price:60,stock:100},{name:"Curd
(kg)",price:80,stock:50},{name:"Cova Sweet (kg)",price:420,stock:20},{name:"Butter
(kg)",price:500,stock:15},{name:"Ghee (litre)",price:650,stock:25}];
const
get=k=>JSON.parse(localStorage.getItem("km_"+k)||"null"),set=(k,v)=>localStorage.setItem("km_"+k,J
SON.stringify(v));
if(!get("products"))set("products",products);["orders","vendors","cows"].forEach(k=>{if(!get(k))set(k,[]
)});
let admin=false;function say(t){document.getElementById("msg").textContent=t}
function show(id){document.querySelectorAll("main
section").forEach(s=>s.classList.add("hidden"));document.getElementById(id).classList.remove("hidde
n");render()}
function render(){let
p=get("products");document.getElementById("productList").innerHTML=p.map(x=>`<article
class="card"><h3>${x.name}</h3><p>₹${x.price}</p><p>Available stock:
${x.stock}</p></article>`).join("");document.getElementById("featured").innerHTML=p.slice(0,3).map
(x=>`<article class="card">${x.name} –
₹${x.price}</article>`).join("");["product","stockProduct"].forEach(id=>document.getElementById(id).i
nnerHTML=p.map(x=>`<option>${x.name}</option>`).join(""));document.getElementById("farmReco
rds").innerHTML="<h3>Vendors</h3>"+get("vendors").map(v=>`<p>${v.name} –
${v.phone}</p>`).join("")+"<h3>Cows</h3>"+get("cows").map(c=>`<p>${c.tag} –
${c.breed}</p>`).join("")}
document.getElementById("adminLogin").onsubmit=e=>{e.preventDefault();if(adminEmail.value==="
admin@komatha.test"&&adminPass.value==="admin123"){admin=true;document.getElementById("ad
minArea").classList.remove("hidden");say("Admin login successful.")}else say("Invalid admin
credentials.")};
```

```
document.getElementById("orderForm").onsubmit=e=>{e.preventDefault();let
p=get("products"),x=p.find(z=>z.name===product.value),n=Number(qty.value);if(!x||n<1||x.stock<n)ret
urn say("Insufficient stock.");x.stock-=n;set("products",p);let
a=get("orders");a.push({customer:customer.value,phone:phone.value,product:x.name,qty:n,total:n*x.pri
ce,date:new
Date().toISOString()});set("orders",a);document.getElementById("orderResult").textContent="Order
placed. Total: ₹"+n*x.price;e.target.reset();render()};
document.getElementById("stockForm").onsubmit=e=>{e.preventDefault();if(!admin)return
say("Admin login required.");let
p=get("products"),x=p.find(z=>z.name===stockProduct.value);x.stock+=Number(stockQty.value);set("
products",p);say("Stock updated.");render()};
function report(period){if(!admin)return say("Admin login required.");let now=new
Date(),a=get("orders").filter(o=>{let d=new Date(o.date);return
period==="day"?d.toDateString()===now.toDateString():period==="month"?d.getMonth()===now.get
Month()&&d.getFullYear()===now.getFullYear():d.getFullYear()===now.getFullYear()});document.ge
tElementById("report").innerHTML=`<p>Orders: ${a.length} | Sales:
₹${a.reduce((s,x)=>s+x.total,0)}</p>`}
document.getElementById("vendorForm").onsubmit=e=>{e.preventDefault();let
a=get("vendors");a.push({name:vendor.value,phone:vendorPhone.value});set("vendors",a);e.target.reset
();render()};
document.getElementById("cowForm").onsubmit=e=>{e.preventDefault();let
a=get("cows");a.push({tag:tag.value,breed:breed.value});set("cows",a);e.target.reset();render()};
render();
</script></body></html>
```

8. Sample Input

Operation	Field	Sample value
Product	Name / Price	Fresh milk / ₹60 per litre
Stock entry	Product / Quantity	Milk / 100 litres
Customer order	Name / Product / Quantity	Ravi / Curd / 2
Vendor	Name / Phone	Lakshmi Dairy / 9876543210
Cow	Tag / Breed	KM-021 / Gir

9. Expected Output with Image

The browser displays the project homepage and navigation options. Users can interact with the forms and view the relevant records or confirmation messages. The exact output depends on the sample data entered during testing.

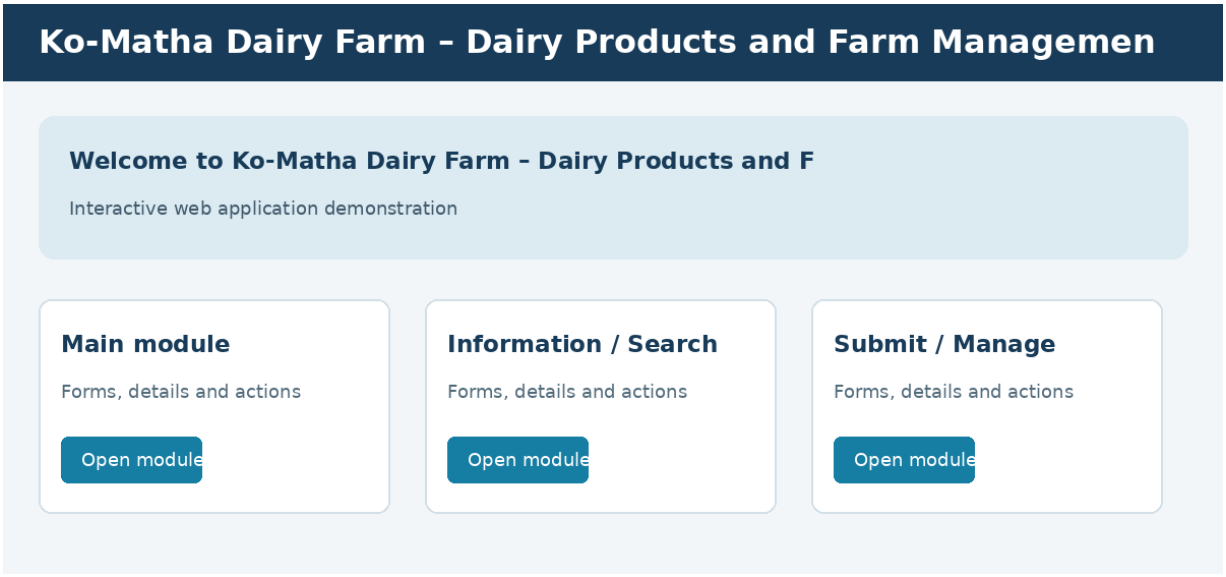


Figure: Illustrative expected output for Experiment 31.

10. Result

Thus, Ko-Matha Dairy Farm – Dairy Products and Farm Management Website was designed and developed successfully using HTML, CSS and JavaScript. The main modules and sample workflows were demonstrated in a web browser.

Product note: product prices and stock are sample values. Any claims about milk type or nutritional properties must be substantiated.

EXPERIMENT 32

1. Title

Ram Infotech – Laptop Sales and Service Management Website

2. Aim

To develop a website for laptop product browsing and sales, service booking and tracking, rental agreement details, customer records and shop stock maintenance.

3. Software Requirements

Component	Requirement
Frontend	HTML5
Styling	CSS3
Programming	JavaScript (ES6)
Editor	Visual Studio Code / text editor
Browser	Google Chrome / Microsoft Edge / Firefox
Storage	Browser LocalStorage (prototype only)
OS	Windows / Linux / macOS

4. Problem Description

Customers need a convenient way to compare laptop models and configurations, purchase devices and request repair services. A shop also needs to track stock, customers, service requests and rental agreements. Ram Infotech combines a sample laptop catalogue, purchase form, service request form, rental form and admin stock/report area.

5. Modules

Admin

- Admin registration / login
- Manage laptop stock
- View sales and service requests
- Manage service information

Sales

- Laptop details
- Model and configuration
- Laptop purchase

Services

- Laptop service information
- Book repair/service
- Track service status

Customer

- Customer information
- Purchase details
- Service details

Rental

- Laptop rental agreement details

Stock

- Add stock
- View available stock

6. Algorithm

49. Initialize sample laptop catalogue and empty customer transaction records.
50. Display laptop models, specifications and prices.
51. Customer submits purchase details; validate stock and save the order.
52. Customer submits a service request with device and problem details.
53. Customer may submit a rental request and rental period.
54. Admin logs in to add stock and review purchases, services and rental requests.
55. Display confirmation/reference numbers for submitted requests.
56. End.

7. Complete HTML, CSS and JavaScript Code

```
<!DOCTYPE html><html lang="en"><head><meta charset="UTF-8"><meta name="viewport"
content="width=device-width,initial-scale=1"><title>Ram Infotech</title><style>
body{font-
family:Arial;margin:0;background:#f3f6fa;color:#273747}header,footer{background:#173b59;color:whi
te;padding:18px}nav{display:flex;flex-
wrap:wrap;gap:5px}button{background:#167ba4;color:white;border:0;padding:10px;border-
radius:5px;cursor:pointer}.wrap{max-width:1050px;margin:20px auto;padding:0
16px}.panel,.card{background:white;border:1px solid #dce5eb;border-
radius:10px;padding:18px;margin:12px 0}.grid{display:grid;grid-template-columns:repeat(auto-
fit,minmax(220px,1fr));gap:12px}.hidden{display:none}input,select,textarea{width:100%;padding:10px
;margin:7px 0 12px;box-sizing:border-box}label{font-weight:bold}
</style></head><body><header><h1>💻 Ram Infotech</h1><nav><button
onclick="show('home')">Home</button><button onclick="show('store')">Laptop
Store</button><button onclick="show('service')">Service Center</button><button
onclick="show('rental')">Rental</button><button
onclick="show('admin')">Admin</button></nav></header><main class="wrap"><div id="msg"></div>
<section id="home" class="panel"><h2>Laptop Sales & Service</h2><p>Explore laptops, request
repairs and enquire about rentals.</p><div id="featured" class="grid"></div></section>
<section id="store" class="panel hidden"><h2>Online Store</h2><div id="catalogue"
class="grid"></div><h3>Purchase Laptop</h3><form id="purchase"><label>Customer
name</label><input id="buyer" required><label>Email</label><input id="buyerEmail" type="email"
required><label>Laptop model</label><select id="model"></select><button>Place Purchase
Order</button></form><p id="purchaseResult"></p></section>
<section id="service" class="panel hidden"><h2>Service Center</h2><p>Services: hardware diagnosis,
OS installation, keyboard replacement, battery and display repair.</p><form
id="serviceForm"><label>Customer name</label><input id="serviceName"
```

```
required><label>Email</label><input id="serviceEmail" type="email" required><label>Laptop
model</label><input id="serviceModel" required><label>Problem description</label><textarea
id="problem" required></textarea><button>Book Service</button></form><div
id="serviceStatus"></div></section>
<section id="rental" class="panel hidden"><h2>Laptop Rental Agreement</h2><form
id="rentalForm"><label>Customer name</label><input id="renter"
required><label>Email</label><input id="renterEmail" type="email" required><label>Laptop
model</label><select id="rentalModel"></select><label>Rental days</label><input id="days"
type="number" min="1" required><button>Submit Rental Request</button></form><div
id="rentalResult"></div></section>
<section id="admin" class="panel hidden"><h2>Admin Login & Stock Maintenance</h2><form
id="adminLogin"><label>Email</label><input id="adminEmail" type="email"
required><label>Password</label><input id="adminPassword" type="password"
required><button>Login</button></form><div id="adminArea" class="hidden"><h3>Add
Stock</h3><form id="stockForm"><label>Model</label><select
id="stockModel"></select><label>Quantity</label><input id="stockQty" type="number" min="1"
required><button>Update Stock</button></form><h3>Recent Requests</h3><div
id="requests"></div></div></section>
</main><footer>Ram Infotech – Lab Prototype</footer><script>
const initial=[{model:"ThinkBook 14",config:"Core i5, 16GB RAM, 512GB
SSD",price:55000,stock:5},{model:"IdeaPad Slim 3",config:"Ryzen 5, 8GB RAM, 512GB
SSD",price:42000,stock:4},{model:"VivoBook 15",config:"Core i3, 8GB RAM, 512GB
SSD",price:38000,stock:3}];
const
get=k=>JSON.parse(localStorage.getItem("ram_"+k)||"null"),set=(k,v)=>localStorage.setItem("ram_"+k
,JSON.stringify(v));
if(!get("laptops"))set("laptops",initial);["sales","services","rentals"].forEach(k=>{if(!get(k))set(k,[])});
let admin=false;function say(t){document.getElementById("msg").textContent=t}
function show(id){document.querySelectorAll("main
section").forEach(s=>s.classList.add("hidden"));document.getElementById(id).classList.remove("hidde
n");render()}
function render(){let
a=get("laptops");document.getElementById("catalogue").innerHTML=a.map(x=>`<article
class="card"><h3>${x.model}</h3><p>${x.config}</p><b>₹${x.price.toLocaleString("en-
IN")}</b><p>Stock:
${x.stock}</p></article>`).join("");document.getElementById("featured").innerHTML=a.slice(0,2).map
(x=>`<article class="card">${x.model} – ₹${x.price.toLocaleString("en-
IN")}</article>`).join("");["model","rentalModel","stockModel"].forEach(id=>document.getElementByI
d(id).innerHTML=a.map(x=>`<option>${x.model}</option>`).join(""));document.getElementById("req
uests").innerHTML="<h4>Sales</h4>"+get("sales").map(x=>`<p>${x.id}: ${x.customer} –
${x.model}</p>`).join("")+"<h4>Services</h4>"+get("services").map(x=>`<p>${x.id}: ${x.customer}
– ${x.problem}</p>`).join("")+"<h4>Rentals</h4>"+get("rentals").map(x=>`<p>${x.id}:
${x.customer} – ${x.model}, ${x.days} days</p>`).join("")}
```

```
document.getElementById("purchase").onsubmit=e=>{e.preventDefault();let
a=get("laptops"),x=a.find(z=>z.model===model.value);if(x.stock<1)return say("Out of stock.");x.stock-
-;set("laptops",a);let
s=get("sales"),id="SALE"+Date.now();s.push({id,customer:buyer.value,email:buyerEmail.value,model:
x.model});set("sales",s);document.getElementById("purchaseResult").textContent="Order received.
Reference: "+id;e.target.reset();render()};
document.getElementById("serviceForm").onsubmit=e=>{e.preventDefault();let
a=get("services"),id="SRV"+Date.now();a.push({id,customer:serviceName.value,email:serviceEmail.va
lue,model:serviceModel.value,problem:problem.value,status:"Received"});set("services",a);document.g
etElementById("serviceStatus").textContent="Service request created: "+id+"
(Received)";e.target.reset()};
document.getElementById("rentalForm").onsubmit=e=>{e.preventDefault();let
a=get("rentals"),id="RNT"+Date.now();a.push({id,customer:renter.value,email:renterEmail.value,model
:rentalModel.value,days:days.value,status:"Pending"});set("rentals",a);document.getElementById("renta
lResult").textContent="Rental enquiry submitted: "+id+". A rental agreement must be reviewed by the
shop.";e.target.reset()};
document.getElementById("adminLogin").onsubmit=e=>{e.preventDefault();if(adminEmail.value==="
admin@raminfotech.test"&&adminPassword.value==="admin123"){admin=true;document.getElement
ById("adminArea").classList.remove("hidden");say("Admin login successful.");render()}else
say("Invalid admin credentials.")};
document.getElementById("stockForm").onsubmit=e=>{e.preventDefault();if(!admin)return
say("Admin login required.");let
a=get("laptops"),x=a.find(z=>z.model===stockModel.value);x.stock+=Number(stockQty.value);set("la
ptops",a);say("Stock updated.");render()};
render();
</script></body></html>
```

8. Sample Input

Operation	Field	Sample value
Laptop	Model / Configuration / Price	ThinkBook 14 / Core i5, 16GB RAM, 512GB SSD / ₹55,000
Purchase	Customer / Model	Anil / ThinkBook 14
Service	Device / Problem	Laptop / Battery not charging
Rental	Model / Period	ThinkBook 14 / 7 days
Admin	Email / Password	admin@raminfotech.test / admin123

9. Expected Output with Image

The browser displays the project homepage and navigation options. Users can interact with the forms and view the relevant records or confirmation messages. The exact output depends on the sample data entered during testing.

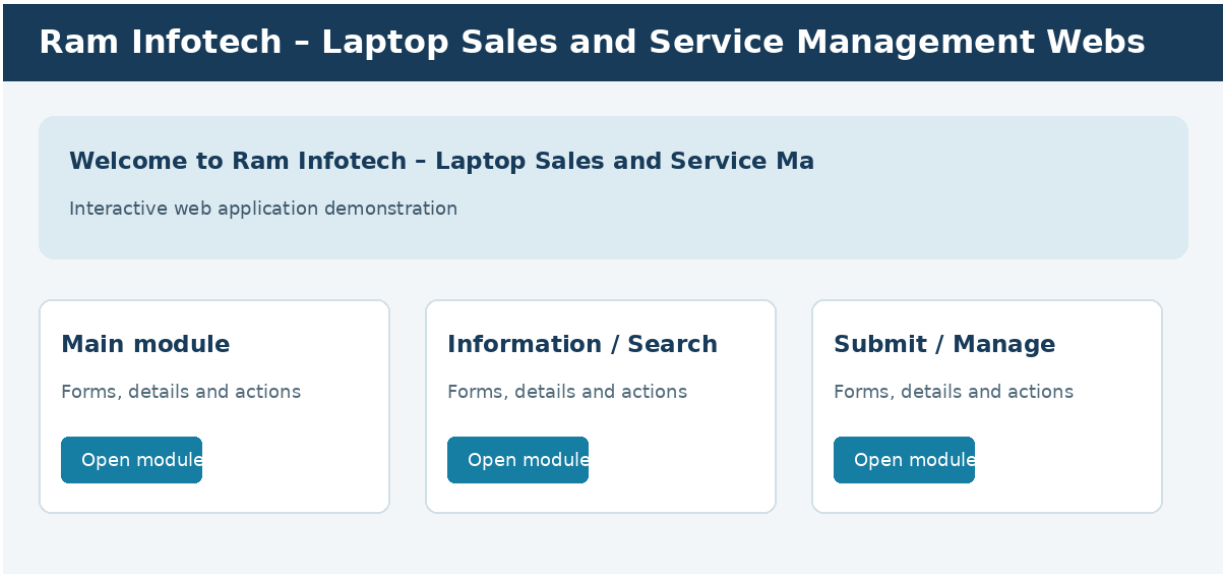


Figure: Illustrative expected output for Experiment 32.

10. Result

Thus, Ram Infotech – Laptop Sales and Service Management Website was designed and developed successfully using HTML, CSS and JavaScript. The main modules and sample workflows were demonstrated in a web browser.